Combinatorial theory esp. related to Schubert polynomials.

PIPEDREAMS (these are SLM), relations to crystal bases

Dan Bump

<u>Lecture 1</u> Schubert calculus. (cf. Fulton, Young Tableaux)

Motivation for Schubert polynomials.

$Gr(n, N)$ = grassmannian of n dim'l subspaces of $\mathbb{C}^N$
$n < N$

$Fl(n)$ = Flag variety $= \{n\text{-flags} \subset \mathbb{C}^n\}$
$V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n = \mathbb{C}^n$

$Gr(n, N) = GL(N, \mathbb{C})/P$, $P = \begin{pmatrix} A & * \\ 0 & D \end{pmatrix}$ $A \in GL(n)$, $D \in GL(k)$ $N = n+k$

$Fl(n) = GL(n)/B$, $B = \begin{pmatrix} * & & * \\ & \ddots & \\ 0 & & * \end{pmatrix}$

<u>Borel</u> 1953 $H^*(Gr(n, N)) \simeq \mathbb{C}[x_1, \cdots x_n]^{\mathfrak{S}_n}/\text{ideal}$

$H^*(Fl(n)) \simeq \mathbb{C}[x_1, \cdots x_n]^{\mathfrak{S}_n}/\text{ideal}$

These varieties have nat'l bases. Schubert varieties $\longleftrightarrow$ permutations or partitions

$G = GL(n, \mathbb{C})$, $Fl(n) = G/B$, $W = \mathfrak{S}_n$ Weyl group

$G = \coprod_{w \in W} BwB$, $X^\circ(w) = BwB/B \simeq \mathbb{C}^{\ell(w)}$, $\ell(w)$ = length of w

$\mathfrak{S}_n = W = \langle s_1, \cdots s_{n-1} \rangle$, $s_i = (i, i+1)$, $\ell(w) =$ shortest k $w = s_{i_1} \cdots s_{i_k}$.

W has a partial order. $y, w \in W$, $w = s_{i_1} \cdots s_{i_k}$.

write $y \leq w$ if $y = s_{j_1} \cdots s_{j_\ell}$, $(j_1, \cdots, j_\ell)$ is a subsequence of $(i_1, \cdots, i_k)$

This def'n does not depend of reduced word for w.

$X(w) := \overline{X^0(w)} = \bigcup_{y \leq w} X^0(y)$, $X^0(w)$ is an open cell inside $X(w)$

Schubert varieties

$Fl(n) = \bigcup X^0(w) = X(w_0)$, $w_0 =$ longest elt.

$\dim_{\mathbb{R}} X^0(w) = 2\ell(w)$. The cellular decomp. implies all $H^i(Fl(n)) = 0$ for i not even

Cup product is commutative.

$X(w)$ is an alg var.

Another description of H^* (Borel)

$Fl(n) = \{V_0 \subsetneq V_1 \subsetneq \cdots \subsetneq V_n\}$

V_i / V_{i+1} is a line bundle on $Fl(n)$

$c_1(V_i / V_{i+1}) = x_i \in H^2$. Generators.

Borel proved $H^*(\cdots) = \mathbb{C}[x_1, \cdots x_n] \Big/ \left(\begin{array}{l}\text{ideal gen'd by} \\ \text{sym. poly. w/} \\ \text{zero c.t.}\end{array}\right)$

<u>Problem</u> What is the multiplicative str. like for Schubert classes?

Lascoux and Schützenberger defined polynomials

$\mathfrak{S}_w = \mathbb{C}[x_1, \dots x_n]$ representing the Schubert classes

Interesting combinatorial theory.

$X(w) = \overline{BwB/B}$

$\Omega(w) = \overline{B^- wB/B}$ codim $\ell(w)$

$[\Omega(w)] \in H^{2\ell(w)}(Fl)$

$\updownarrow$

$\mathfrak{S}_w$

$N = n+k$. $\quad Gr(n,N)$ = n-dim'l subspaces of $GL(N,\mathbb{C})$

$\simeq G/P$, $\quad P = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \begin{matrix} n \\ k \end{matrix}$

The intersection theory here goes back to Schubert, Giambelli, Pieri, ...

Lascoux - Schützenberger

$$G = \bigcup_{w \in W} Bw^{-1}P$$

$w(1) < w(2) < \dots < w(n)$

$w(n+1) < w(n+2) < \dots < w(N)$

w has a unique descent (at n)

"Grassmann permutation"

Grassmann perms coset reps for $(\mathfrak{S}_n \times \mathfrak{S}_k) \backslash \mathfrak{S}_N$

Schubert cells are $\overline{Bw^{-1}P/P}$

Dual Schubert cells $\overline{B^- w^{-1} P/P} = \Omega(w)$ dual cells

Intersection theory for Schubert cells understood by Pieri, Giambelli

Schur Polynomials

$\lambda = (\lambda_1, \dots \lambda_n)$ is a partition $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$

$$s_\lambda(x_1, \dots x_n) = \frac{\det(x_i^{\lambda_j + n - j})}{\det(x_i^{n-j})}$$

By WCF, $\exists$ irrep $\Pi_\lambda^{GL(n)}$ of $GL(n,\mathbb{C})$. $\chi_\lambda(g) = s_\lambda$ (eigen values)

$$s_\lambda s_\mu = \sum_\gamma c_{\lambda\mu}^{\gamma} s_\gamma$$ ← C-R coeff

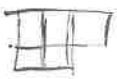
if $\mu = (k) = (k,0,\cdots,0)$, $s_{(k)} = h_k = \sum_{i_1 \le \cdots \le i_k} x_{i_1} \cdots x_{i_k}$: complete sym poly

$$\Pi_{(k)}^{GL(n)} = \mathrm{Sym}^k \mathbb{C}^n$$

Pieri's formula $s_\lambda s_{(k)} = \sum_\gamma s_\gamma$

γ/λ is a horizontal strip of length k

Condition : $YD(\nu)$ = boxes at (i,j) w/ $j \le \lambda_i$

$YD(3,2) =$ 

$YD(\nu) \supset YD(\lambda)$, $YD(\gamma) - YD(\lambda)$ has no 2 boxes in same column $\rightsquigarrow$ horizontal strip

Jacobi - Trudi identity

$$s_\lambda = \det \begin{pmatrix} h_{\lambda_1} & h_{\lambda_1+1} & h_{\lambda_1+2} & \cdots \\ h_{\lambda_2-1} & h_{\lambda_2} & h_{\lambda_2+1} & \cdots \\ \vdots & \vdots & \vdots & \end{pmatrix}$$

$h_i = 0$ if $i < 0$

$h_0 = 1$

Pieri, Giambelli

↑ found formulas in $H^*(Fl)$

GIAMBELLI = JTI

These formulas are same:

<u>Lemma</u>

$w(1) < \cdots < w(n)$, $w(n+1) < \cdots < w(N)$

are in bijection w/ partitions $\lambda \subset n \times k$ rectangles.

$1 \le i \le n$, $\lambda_i = k + i - w(i)$

$w(i) = k + i - \lambda_i$

$[\sigma_w]$ Schubert class $\longleftrightarrow$ s_λ

$\uparrow$

$H^*(Gr(n, N))$

$H^*(Gr(n, N)) \simeq \mathbb{C}[x_1, \cdots, x_n]^{\mathfrak{S}_n} / \text{ideal} = \mathbb{C}\text{-span of } s_\lambda$ s.t.

$YD(\lambda) \subset n \times k$ rectangle

L & S: simila picture for flag varieties.

<u>Lecture 2</u> Demazure operators and Demazure characters

Demazure ops were introduced independently by Bernstein – Gelfand – Gelfand and Demazure (early 1970's) in connect w/ Schubert geometry.

Let $\mathfrak{S}_n$ act on $\overset{R}{\overset{\|}{\mathbb{C}[x_1, \cdots, x_n]}}$ by permuting variables. $f \in R$, then f acts by multiplication. $\mathfrak{S}_n$, R both are in $\mathrm{End}(R)$

$wfw^{-1} = {}^{w}f$.

Lemma If $f \in R$, then $\dfrac{f - s_i f}{x_i - x_{i+1}} \in R$ $\quad s_i = (i, i+1)$

Pf Numerator vanishes when $x_i = x_{i+1}$, so zeroes of numerator and denominator cancel.

The operator $D_i = (x_i - x_{i+1})^{-1}(1 - s_i)$ maps $f \mapsto \dfrac{f - s_i f}{x_i - x_{i+1}}$

These divided difference ops are one kind of Demazure op.

$g = D_i f$ satisfies $s_i g = g$ $\quad$ (symmetric in $x_i \longleftrightarrow x_{i+1}$)

If f is symmetric in $x_i \longleftrightarrow x_{i+1}$, then $D_i f = 0$, therefore $D_i^2 = 0$.

Also D_i satisfy braid relations: $\quad D_i D_j = D_j D_i$ if $|i-j| > 1$

$$D_i D_{i+1} D_i = D_{i+1} D_i D_{i+1}.$$

Pf of second relation: $\quad D_1 D_2 D_1 = D_2 D_1 D_2$

$$\text{LHS} = (x_1 - x_2)^{-1}(1 - s_1)(x_2 - x_3)^{-1}(1 - s_2)(x_1 - x_2)^{-1}(1 - s_1)$$

$$= (x_1 - x_2)^{-1}(x_1 - x_3)^{-1}(x_2 - x_3)^{-1} \sum_{w \in \mathfrak{S}_3} \operatorname{sgn}(w) \cdot w$$

$D_2 D_1 D_2$ has same evaluation

The braid relation is some close relative of Yang-Baxter equation.

Artin braid group $\mathbb{B}_{n-1}$

$\mu_1, \dots, \mu_{n-1}$ subject to braid rel'ns $\quad \mu_i \mu_{i+1} \mu_i = \mu_{i+1} \mu_i \mu_{i+1}$

$\mu_i \mu_j = \mu_j \mu_i$ if $|i-j| > 1$

Symmetric group is a Coxeter group.

s_i also satisfy braid relations and also a quadratic rel'n $s_i^2 = 1$.

Nil Hecke algebra (Demazure ops)

braid rel'ns

$D_i^2 = 0$.

If q is an indeterminate, gens T_i

braid rel'n

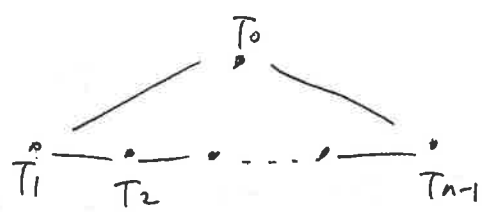
$T_i^2 = (q-1)T_i + q$

This is the $GL(n)$ HA

Iwahori – Matsumoto, convolution alg of $GL(n, F)$, $F = \mathbb{Q}_p$, $q = p$

T_i = char func. on $J s_i J$, J = Iwahori

$0 \le i \le n-1$



Lusztig (1978) The same Hecke alg can be realized as a ring acting on equiv. K-theory of complex flag variety.

<u>Matsumoto's thm</u>. relation between diff. reduced words for elts of S_n (or Coxeter groups)

$w = s_{i_1} \cdots s_{i_k}$ reduced if k is minimal

<u>Thm</u> If $s_{i_1} \cdots s_{i_k} = s_{j_1} \cdots s_{j_k}$ are two reduced expressions, then

their equality can be proved using only braid relations.

<u>Example</u> in $\mathfrak{S}_4$,

$$w_0 = s_1 s_2 s_1 s_3 s_2 s_1 = s_3 s_2 s_3 s_1 s_2 s_3$$

$\ell(w_0) = 6$

1 2 1 3 2 1

1 2 3 1 2 1

1 2 3 2 1 2

1 3 2 3 1 2

⋮

Conclusion really means $\mu_{i_1} \cdots \mu_{i_k} = \mu_{j_1} \cdots \mu_{j_k}$ in $\mathbb{B}_{n-1}$

<u>Application to nil Hecke alg.</u>

D_i $\left[\begin{array}{l} \text{braid rel'n} \\ D_i^2 = 0 \end{array} \right.$

If $w \in \mathfrak{S}_n$, let $w = s_{i_1} \cdots s_{i_k}$ (reduced) then we can define

$D_w = D_{i_1} \cdots D_{i_k}$ Demazure operators

well-def'd by Matsumoto's thm since D_i satisfy braid rel'ns.

<u>Schubert polynomials</u> $\Lambda = \mathbb{Z}^n$ ($GL(n)$ weight lattices)

If $\lambda = (\lambda_1, \cdots, \lambda_n) \in \Lambda$, $x^\lambda = x_1^{\lambda_1} \cdots x_n^{\lambda_n}$. $\rho = (n-1, n-2, \cdots, 0)$ = Weyl vector

$s_i \rho = \rho - \alpha_i$, $\alpha_i = (0, \cdots, 0, \underset{\substack{\uparrow \\ i\text{th}}}{1}, -1, 0, \cdots)$

<u>Def'n</u>. $S_w(x) = D_{w^{-1} w_0} \cdot x^\rho$

Schubert poly.

Lecture 3

G cplx analytic Lie gp, reductive

T max'l torus

$\Lambda = X^*(T)$ = gp of rat'l chars.

$\Phi \subset \Lambda$ root system

work in a ring R containing W and $\mathcal{O}(T)$.

$wfw^{-1} = {}^w f$

$$\partial_i := (1 - z^{-\alpha_i})^{-1}(1 - z^{-\alpha_i} s_i)$$

$$\partial_i f(z) = \frac{f(z) - z^{-\alpha_i} \cdot f(s_i z)}{1 - z^{-\alpha_i}}$$ "isobaric Demazure operators"

$\boxed{\partial_i z^\lambda}$

$$\partial_i z^\lambda = \frac{z^\lambda - z^{-\alpha_i} z^{\lambda - \langle \alpha_i^\vee, \lambda \rangle \alpha_i}}{1 - z^{-\alpha_i}}$$

$$= z^\lambda \frac{1 - z^{-\alpha_i}(\langle \alpha_i^\vee, \lambda \rangle + 1)}{1 - z^{-\alpha_i}}$$

This is a finite geometric series.

If $\langle \alpha_i^\vee, \lambda \rangle = k \geq 0$, this equals $z^\lambda (1 + z^{-\alpha_i} + z^{-2\alpha_i} + \cdots + z^{-\langle \lambda, \alpha_i^\vee \rangle \alpha_i})$

$= z^\lambda + z^{\lambda - \alpha_i} + \cdots + z^{s_i \lambda}$

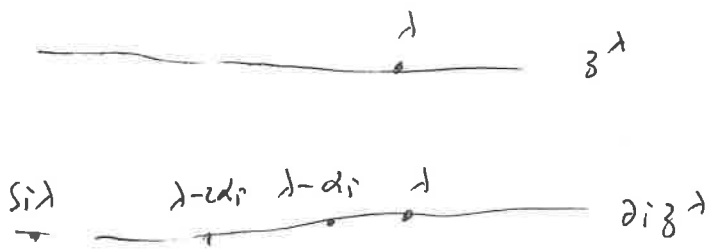
s_i - invariant

If $\langle \alpha_i^\vee, \lambda \rangle = -1$, $\partial_i z^\lambda = 0$.

If $\langle \alpha_i^\vee, \lambda \rangle < -1$, this is $-(z^{\lambda + \alpha_i} + z^{\lambda + 2\alpha_i} + \cdots + z^{s_i \lambda - \alpha_i})$

still s_i- invariant

For $GL(2)$:



∂_i symmetrizes. Can check directly $s_i \partial_i = \partial_i$.

<u>Thm</u> $\partial_i^2 = \partial_i$ and they satisfy braid relations.

Therefore we may apply Matsumoto's theorem and define

$\partial_w = \partial_{i_1} \cdots \partial_{i_k}$, $s_{i_1} \cdots s_{i_k} = w$ reduced expr.

<u>Thm</u> Let λ be dominant. then $\partial_{w_0} z^\lambda$ = char. of h.w. rep of h.w. λ.

In general, $\partial_w z^\lambda$ (λ dominant) are called Demazure characters or key polynomials.

There is a refinement (Littlemann, Kashiwara) of $\partial_{w_0} z^\lambda = s_\lambda$ involving crystals.

<u>Lecture 4</u>. Tokuyama models. Involve a parameter q? $\sim$ fermionic models for Schur poly.

$q=0$ interesting

$q=1$ interesting

<u>History</u>; free-fermionic 6-vertex model was studied by Felderhof, Korepin

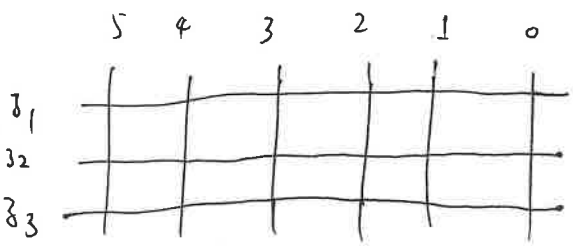
Particular models in free-fermionic case have Schur poly. or more general

$$(*) \quad \underbrace{\prod_{i<j} (z_i - q z_j)}_{\text{deformed Weyl denominator}} \; s_\lambda(z_1, \dots, z_n) \qquad \text{as partition functions.}$$

This is a combinatorial fact equiv. to a formula of Tokuyama expressing $(*)$ as a sum of Gelfand-Tsetlin patterns. It was translated into a statement about lattice models by Hamel and King. Brubaker, Bump Friedberg. Paul Zinn-Justin independently introduce Yang-Baxter approach to Tokuyama's formula. Relevant YBE is in earlier work of Korepin.

What are the models:

Grid w/ n rows and N columns, $N \geq \lambda_1 + n - 1$



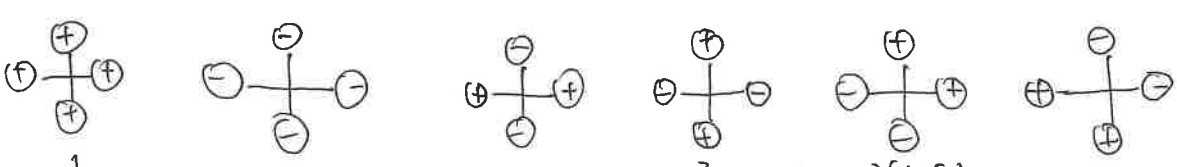
rows are assoc. w/ parameters $z_1, \dots, z_n$

In a state of the model every edge is assigned a "spin" $\oplus$ $\ominus$

Spins on boundary edges are fixed.

Put $\ominus$ on right boundary, top boundary at locations $\lambda_i + n - i$.

In a state of the model, $\lambda = (\lambda_1, \dots, \lambda_n)$ a partition



Each state $s \rightsquigarrow$ Boltzmann weight $\beta(s) = \prod_{v \text{ vertex}} \beta_v(s)$

$$Z(S) = \sum_{\text{state } s} \beta(s) \qquad \text{partition function.}$$

$\uparrow$ model

In statistical mechanics, $\dfrac{\beta(s)}{Z(S)}$ = probability of s.

<u>Thm</u> $Z(S) = \prod_{i<j} (z_i - q z_j)\, s_\lambda(z)$

Schur polynomials are chars of irreps of $GL(n, \mathbb{C})$

$$\chi_\lambda^{GL(n)}(g) = s_\lambda(z_1, \cdots, z_n), \qquad z_1, \cdots, z_n \text{ eigenvalues of } g$$

There are two formulae for s_λ:

$$s_\lambda(z_1, \cdots, z_n) = \frac{\det\left(z_i^{n+\lambda_i - i}\right)}{\det\left(z_i^{n-i}\right)}$$

denominator $\begin{pmatrix} z_1^2 & z_2^2 & z_3^2 \\ z_1 & z_2 & z_3 \\ 1 & 1 & 1 \end{pmatrix} = \prod_{i<j}(z_i - z_j)$ Vandermonde det.

numerator is an alternating polynomial.

$$\begin{vmatrix} z_1^{\lambda_1+2} & z_2^{\lambda_1+2} & z_3^{\lambda_1+2} \\ z_1^{\lambda_2+1} & z_2^{\lambda_2+1} & z_3^{\lambda_2+1} \\ z_1^{\lambda_3} & z_2^{\lambda_3} & z_3^{\lambda_3} \end{vmatrix}$$

$\text{num}(s_i \cdot z) = -\text{num}(z)$

$\Rightarrow \dfrac{\det\left(z_i^{n+\lambda_i - i}\right)}{\det\left(z_i^{n-i}\right)}$ is a symmetric poly.

Jacobi's def'n of Schur poly.

<u>The other def'n</u> Let T be a semistandard Young tableau of shape λ.

$wt(T) = (\mu_1, \mu_2, \cdots)$, $\mu_i = \#$ of i's in T

$$wt\left(\begin{array}{|c|c|c|}\hline 1 & 1 & 2 \\ \hline 3 \\ \cline{1-1}\end{array}\right) = (2,1,1)$$

<u>Def</u> (Littlewood) $\quad s_\lambda(z_1, \cdots, z_n) = \sum_T z^{wt(T)}$

<u>Q</u>. Why is this symmetric ?

Why is this equiv. to Jacobi's def ?

Both questions answered by Tokuyama ~~Bracket~~.

After proving $\dfrac{\Sigma(s_\lambda, q)}{\prod_{i<j}(z_i - q z_j)}$ is a symmetric poly. indep. of q

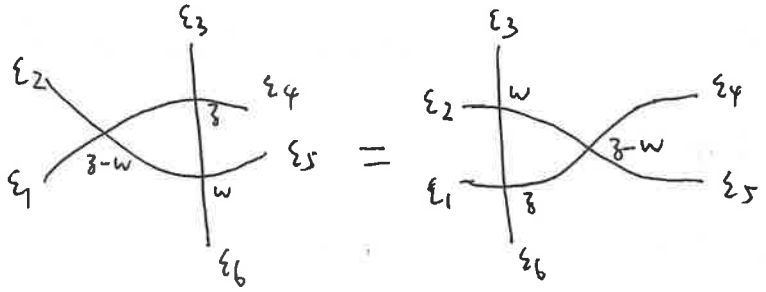
Specialize: $\quad q = 1 \rightsquigarrow$ Jacobi's def'n

$q = 0 \rightsquigarrow$ combinatorial def'n.

$U_q(sl(1|1))$ Super alg Q.G.

$U_q(sl_2)$

<u>Lecture 5</u> Yang - Baxter equation.



Sum over spins on interior edges

λ a partition, $\lambda = (\lambda_1, \dots, \lambda_n)$

Put $\ominus$ spins on right boundary, $\oplus$ on left and bottom.

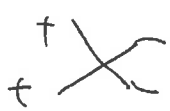
and at top $\ominus$ in columns $\lambda_i + n - i$.

<u>Theorem</u>. There is a symmetric polynomial $S_\lambda(x_1, \dots x_n)$ s.t.

$$Z(S_\lambda) = \prod_{i<j} (z_i - q z_j)\, S_\lambda(x_1, \dots x_n)$$

<u>Lemma</u> $(z_{i+1} - q z_i)\, Z(z_1, \dots z_n; q) = (z_i - q z_{i+1})\, Z(s_i \underline{z}; q)$

<u>Proof</u> Attach R-matrix between $i, i+1$ rows.



$Z(z, q)$ is a poly in $z_1, \dots, z_n, q$

$$\frac{Z(z, q)}{\prod_{i<j} (z_i - q z_j)}$$

<u>Claim</u>. This is W-invariant, $W = \mathfrak{S}_n = \langle s_1, \dots s_{n-1} \rangle$

so $Z(S_\lambda) = \prod_{i<j} (z_i - q z_j)\, S_\lambda(z; q)$ where $S_\lambda(z; q)$ is symmetric in z_i.

Want to show $S_\lambda(z; q)$ is poly. indep of q.

It is indep of q since numerator & denominator are all poly of deg $\frac{1}{2} n(n-1)$ in q

of $\ominus$ spins on vertical edges in any state.

Lecture 6. Tokugame model $Z(S_\lambda(z;q)) = \prod_{i<j}(z_i - q z_j) \cdot S_\lambda(z_1, \dots, z_n)$

$q = 1$, leads to $n! = |W|$ states

$$S_\lambda(z) = \frac{\sum_w \cdots}{\prod_{i<j}(z_i - z_j)}$$ usual Schur

$q=0$. states are in bijection w/ Gelfand-Tsetlin patterns

and SSYT (semi-standard Young tableaux)

$\rightsquigarrow$ combinatorial def'n of Schur function.

A Gelfand-Tsetlin pattern of size n and shape λ

rows indexed by $0, 1, 2, \cdots$

0-th row is λ

Each row is a partition $(a_{i,1}, \dots, a_{i,n-i})$

Rows interlace

$$\lambda = (\lambda_1, \dots, \lambda_n)$$

$$\mu = (\mu_1, \dots, \mu_{n-1})$$

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \cdots \geq \lambda_n$$

$n = 3$

$\lambda_1 \quad \lambda_2 \quad \lambda_3$

$a \quad b$

c

$\lambda_1 \geq a \geq \lambda_2 \geq b \geq \lambda_3$

$a \geq c \geq b$

$GL(n) \rightsquigarrow GL(n-1)$ branching rule

$\lambda = (\lambda_1, \cdots \lambda_n)$ a partition, $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$

There is a irrep of $GL(n)$ of h.w. λ

<u>Thm</u> $\operatorname{Dim} \pi_\lambda^{GL(n)} = \#$ of GTP w/ top row λ

Follows from branching rule $GL(n-1) \to GL(n)$

$$g \mapsto \begin{pmatrix} g & \\ & 1 \end{pmatrix}$$

$$\pi_\lambda^{GL(n)}\Big|_{GL(n-1)} = \bigoplus_{\substack{\mu = (\mu_1, \cdots, \mu_{n-1}) \\ \lambda, \mu \text{ interlace}}} \pi_\mu^{GL(n-1)}$$

GTP for λ read off μ from second row

If we discard boxes in a SSYT containing n, obtain a SSYT of shape μ

Some μ interleaving λ.

Given a state of six vertex model, the vertical edges containing $\ominus$ edges

give a GTP (of size $\lambda + \rho$)

strict GTP $\xleftrightarrow{\sim}$ $s_\lambda^*(z;q)$

GTP of size n, top row $\lambda + \rho$

$\updownarrow$ bijection

SSYT of shape $\lambda + \rho$

Strict means $a_{i,1} > a_{i,2} > a_{i,3} > \cdots$

Using reduced GTP,

States $q=0$ $\longleftrightarrow$ GTP top row λ $\longleftrightarrow$ SSYT of shape λ

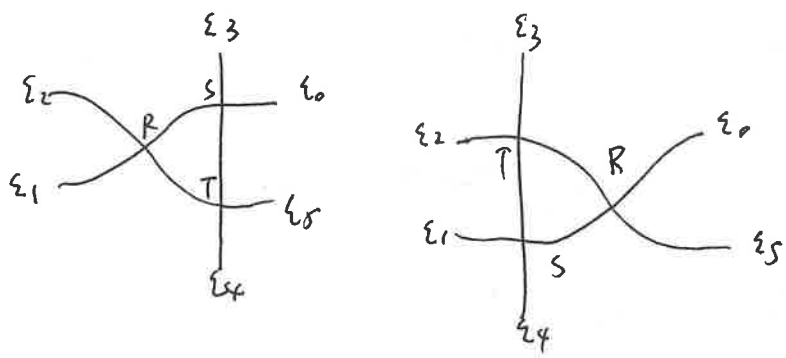
$$\beta(\sigma) = z^{\rho} \cdot z^{w_0 \, wt(T)}$$

$$Z(S_\lambda(z,0)) = z^{\rho} \cdot S_\lambda(z_1, \dots, z_n)$$

$$\text{so } S_\lambda(z) = \sum_{SSYT} z^{w_0 \, wt(T)} \overset{S_\lambda \text{ is symmetric}}{=\!=\!=} \sum_{SSYT} z^{wt(T)}$$

Lecture 7 Parametrized Yang-Baxter Equations.

(1) Equivalence of two systems

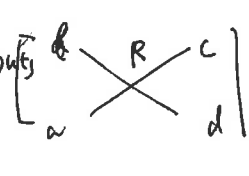


Moving R across the S, T vertices switches . Then used in train argument.

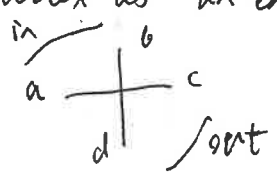
(2) Linear algebra statement:

$\varepsilon_i \in \Sigma$ a set of allowed spins

If V = free vec. sp. on Σ, we can regard a vertex as an endomorphism of $V \otimes V$ as follows: inputs



$a, b, c, d \in \Sigma$



$a \otimes b \in V \otimes V$

$d \otimes c$

$R: V \otimes V \to V \otimes V$ is the linear transf.

$$R(a \otimes b) = \sum \beta\left(\begin{smallmatrix} b & & c \\ & \times & \\ a & & d \end{smallmatrix}\right) d \otimes c$$

$$\langle d \otimes c \mid R \mid a \otimes b \rangle = \beta\left(\begin{smallmatrix} b & & c \\ & \times & \\ a & & d \end{smallmatrix}\right)$$

$\tau: V \otimes V \to V \otimes V, \quad x \otimes y \mapsto y \otimes x$

$\tau \circ R$ is often considered the R-matrix. Both conventions are used.

Yang-Baxter equation is an identity on $V \otimes V \otimes V$

$$(T \otimes I)(I \otimes S)(R \otimes I) = (I \otimes R)(S \otimes I)(I \otimes T)$$

This is equiv to other formulation.

$R, S, T \in \mathrm{End}(V \otimes V)$

$[\![R, S, T]\!]$

$= (T \otimes I)(I \otimes S)(R \otimes I) - (I \otimes R)(S \otimes I)(I \otimes T)$

= Yang-Baxter commutator.

$[\![R, S, T]\!] = 0 \iff$ YBE

Parametrized YBE: Let Γ be a group, usually $\mathbb{C}^\times$, $\mathbb{C}$, elliptic curve.

One exotic example: $GL(2) \times GL(1)$

V some vec. sp. $\quad R: \Gamma \to \mathrm{End}(V \otimes V)$ some map. - Free fermionic six vertex model (Tokuyama)

$[\![R(\gamma), R(\gamma\delta), R(\delta)]\!] = 0$

Variant: homogeneous YBE

$z_1, z_2 \in \mathbb{C}^\times$

$R(r_1, r_2)$

$[R(r_1,r_2), R(r_1,r_3), R(r_2,r_3)] = 0$

<u>Lecture 8</u>. Demazure characters, Demazure crystals. lattice models whose part. func. are Demazure char.

Review: $\partial_i = (1 - z^{-\alpha_i})^{-1}(1 - z^{-\alpha_i} s_i)$

$s_i \partial_i f = \partial_i f$

∂_i satisfy braid relations, so by Matsumoto's thm, we can define ∂_w for $w \in W$.

If $w = w_0$ and λ is dominant, $\partial_{w_0} z^\lambda$ = irr. char. w/ h.w. λ

$= s_\lambda(z)$ Schur function

Combinatorial def'n of Schur poly. (D.E. Littlewood)

$$s_\lambda(z) = \sum_T z^{wt(T)}, \quad T \in SSYT \text{ of shape } \lambda.$$

Kashiwara and Nakashima show the set of SSYT have a structure of a "crystal" w/ remarkable combinatorial properties.

<u>Def.</u> A Kashiwara crystal $\mathcal{C}$ is a directed graph. It comes w/ a map $wt: \mathcal{C} \to \Lambda$ (weight lattice) The edges are labelled by roots.

consider an edge $x \xrightarrow{i} y$ if this edge exists, $wt(y) = wt(x) - \alpha_i$

Kashiwara operators:

$$e_i, f_i : \mathcal{C} \to \mathcal{C} \cup \{0\}$$

0 is an auxiliary symbol connecting failure to apply the operator

$e_i(x) =$ the unique elt z s.t. an edge $\underset{z}{\bullet} \xrightarrow{i} \underset{x}{\bullet}$ exists

$e_i(x) = 0$ if no such edge

$f_i(x) =$ unique y s.t. $\overset{x}{\bullet} \xrightarrow{i} \overset{y}{\bullet}$

$= 0$ if no such edge

$\varepsilon_i, \varphi_i : \mathcal{C} \to \mathbb{Z}$ (even $\mathbb{N}$)

$\varepsilon_i(x) = \max\{k : e_i^k(x) \neq 0\}$ = # of times e_i can be applied

Littelmann and Kashiwara proved a refined Demazure character formula:

The operators ∂_i: $\mathcal{O}(T)$ functions on T $\longrightarrow \mathcal{O}(T)$

$T =$ torus whose ring of reg. functions is Λ

$\mathcal{O}(T) =$ span of z^λ $(\lambda \in \Lambda)$

or $\partial_i : \mathbb{Z}[\Lambda] \to \mathbb{Z}[\Lambda]$

can be lifted to $\mathbb{Z}[B_\lambda] \to \mathbb{Z}[B_\lambda]$, $B_\lambda =$ Kashiwara crystal of SSYT of shape λ.

If $T \in B_\lambda$, suppose $\overset{k}{\overset{\shortparallel}{\langle wt(T), \alpha_i^\vee \rangle}} \geq 0$.

$\partial_i z^\mu = z^\mu + z^{\mu - \alpha_i} + \cdots + z^{\mu - k\alpha_i}$

$\mu = wt(T)$, $\quad \partial_i z^{\mu} = \sum_{j=0}^{k} z^{\mu - j\alpha_i}$

<u>Def'n</u> (Littelmann)

$$\partial_i T = \sum_{j=0}^{k} f_i^j(T) \qquad \text{in } \mathbb{Z}[B_\lambda].$$

<u>Thm</u> (Littelmann) Applied to the h.w. elt these operators braid, so

$$\partial_w T_\lambda = \partial_{i_1} \cdots \partial_{i_r} T_\lambda \quad , w = s_{i_1} \cdots s_{i_r}$$

$\uparrow$ h.w. elt

is independent of the reduced expression.

If $\langle wt(T), \check{\alpha}_i \rangle < 0$,

$$\partial_i(T) = -e_i(T) - e_i^2(T) - \cdots$$

# of applications is $-\langle wt(T), \check{\alpha}_i \rangle - 1$.

<u>Thm</u>. There is a subset of B_λ : $B_\lambda(w)$ s.t. in $\mathbb{Z}[\Lambda]$,

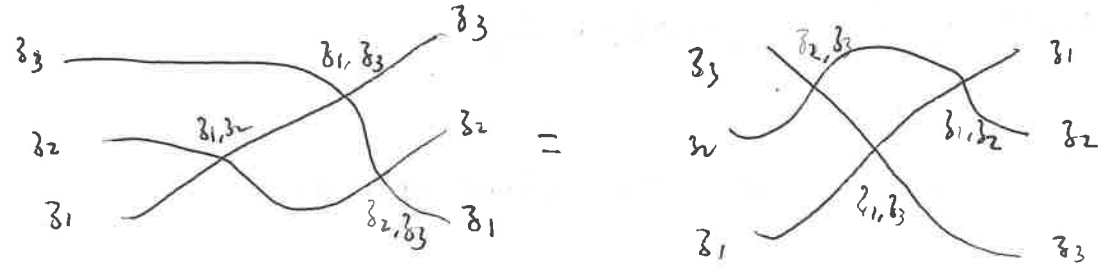
$$\partial_w T_{H.W.} = \sum_{v \in B_\lambda(w)} v.$$

This is called a Demazure crystal

<u>Lecture 8</u>. The R matrix is an endomorphism of $V \otimes V$

$V =$ free vector space on the set of spins.

$$\begin{tikzcd}
 & V\otimes V\otimes V \arrow[dl, "R(z_1,z_2)\otimes I"'] \arrow[dr, "I\otimes R(z_2,z_3)"] & \\
V\otimes V\otimes V \arrow[d, "I\otimes R(z_2,z_3)"'] & & V\otimes V\otimes V \arrow[d, "R(z_1,z_3)\otimes I"] \\
V\otimes V\otimes V \arrow[dr, "R(z_2,z_3)\otimes I"'] & & V\otimes V\otimes V \arrow[dl, "I\otimes R(z_1,z_2)"] \\
 & V\otimes V\otimes V &
\end{tikzcd}$$



On Friday we considered open models. The above formalism gives if $s_i w > w$

$$Z(S_{\lambda, s_i w}) = {}^{\rho}\partial_i^{\circ}\, Z(S_{\lambda, w})$$

$$\partial_i^{\circ} = (z^{\alpha_i} - 1)^{-1}(1 - s_i)$$

$$z^{\rho} D z^{-\rho} = {}^{\rho}D$$

$${}^{\rho}\partial_i^{\circ} = (z^{\alpha_i} - 1)^{-1}(1 - z^{\alpha_i} s_i)$$

By induction $Z(S_{\lambda, w}) = z^{\rho}\, \underbrace{\partial_w^{\circ}\, z^{\lambda}}_{\text{Demazure atom}}$

$$S_{\lambda}(z) = \sum_{w \in W} \partial_w^{\circ}\, z^{\lambda}$$

· This has an analog for the crystal B_λ = crystal of SSYT of shape λ

$$\sum_{w \leq w} z^{wt(T)} = S_\lambda(z)$$

This is a sum over states of the $q=0$ Tokuyama model.

Thm (Brubaker, Buciumas, B, G)

The states of the open model are in bijection w/ a subset $B_\lambda^\circ(w)$ called a Demazure atom.

For closed model, $Z(S_{\lambda,w}) = z^\rho \partial_w z^\lambda$

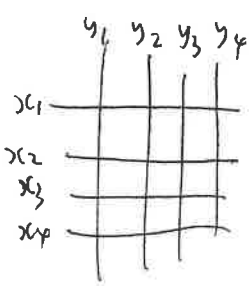
$$\partial_w z^\lambda = \sum_{B_\lambda(w)} z^{wt(T)}$$

$\uparrow$

Demazure crystal

<u>Thm</u> The states of the closed model are in bijection w/ a Demazure crystal.

Parametrized YBE naturally lead to introduction of column parameters



param YBE: Γ a group

$$R : \Gamma \longrightarrow \mathrm{End}(V \otimes V)$$

$$[\![R(z_1, z_2), R(z_1, z_3), R(z_2, z_3)]\!] = 0$$

If $R, S, T \in \mathrm{End}(V \otimes V)$, $[\![R, S, T]\!] = (R \otimes I)(I \otimes S)(T \otimes I) - (I \otimes T)(S \otimes I)(I \otimes R)$

we can choose $x_i \in \Gamma$, $y_i \in \Gamma$

Choose the Boltzmann weights $T_{ij} = R(x_i, y_j)$.

<u>Lecture 9</u>. Can introduce column parameters

$\rightsquigarrow$ factorial Schur functions

double Schubert polynomials

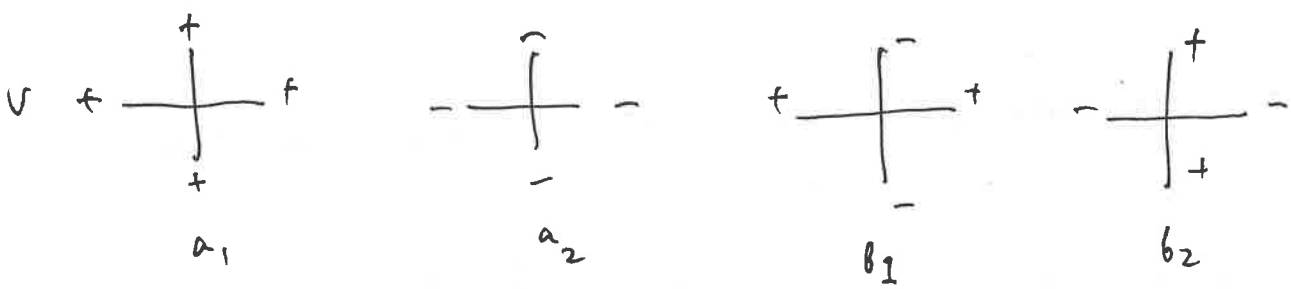
consider when the Boltzmann weights come from a parametrized YBE.

Γ = group, V a vec. sp. $R : \Gamma \to \mathrm{End}(V \otimes V)$ s.t. in $\mathrm{End}(V \otimes V \otimes V)$

$[\![R(\gamma), R(\gamma\delta), R(\delta)]\!] = 0$

<u>Eg 1</u> (Baxter)

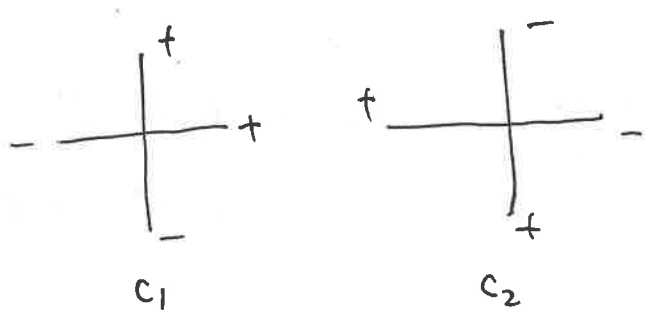
Fix Δ and consider Boltzman weights



"Field-free" $a_1 = a_2 = a$

$b_1 = b_2 = b$

$c_1 = c_2 = c$



$$\Delta(u) = \frac{a^2 + b^2 - c^2}{2ab}$$

<u>Thm (Baxter)</u> If S, T are given, $\Delta(S) = \Delta(T) =: \Delta$ there exists R w/

$[\![R, S, T]\!] = 0$, and $\Delta(R) = \Delta$

We can find q s.t. $\Delta = \frac{1}{2}(q^2 + q^{-2})$

We can find a map

$\longrightarrow$

$\mathfrak{g}$ VM, not assume field free

<u>Example 2</u>. the weights are called free-fermionic if $a_1 a_2 + b_1 b_2 = c_1 c_2$

<u>Thm</u> If S, T are free-fermionic. there is always R s.t. $[\![R, S, T]\!] = 0$.

Moreover, there is a PYBE $\Gamma = GL(2, \mathbb{C}) \times \mathbb{C}^\times$

whose image is exactly the free fermionic case.

Quantum groups:

<u>Eg 1</u>. $U_q(sl_2)$

<u>Eg 2</u>. $U_q(sl(1|1))$

free-fermionic

Tokuyama weights are free-fermionic.

We can choose $z_1, \dots, z_n \in \Gamma$, and ask for an R-mat. that will make train argument.

$\longrightarrow$

Schur polynomials are related to classes of Schubert varieties in

$H^*(\text{Grassmannian}) \simeq$ Symmetric poly. / ideal

The factorial Schur poly. corresp. to equivariant cohomology.

Lecture 10. Schubert poly. were introduced by Lascoux and Schützenberger to describe cohomology of flag varieties.

$x^\rho = x_1^{n-1} x_2^{n-2} \cdots x_{n-1}$ (doesn't include x_n)

Properties: $\mathfrak{S}_{w_0} = x^\rho$

If $s_i w < w$, then $\mathfrak{S}_{s_i w} = D_i \mathfrak{S}_w$, where $D_i = (x_i - x_{i+1})^{-1}(1 - s_i)$

Systems whose partition functions ... Schubert polynomials (pipedreams)

Lecture 11 Stanley's symmetric functions.

1984, Stanley computed the number of reduce expr. for $w = w_0$ in S_n by a method applicable to other w.

Thm There are $\dfrac{\binom{n}{2}!}{1^{n-1} 3^{n-2} 5^{n-3} \cdots (2n-3)^1}$ reduced exp. for w_0

Def. A permutation is <u>decreasing</u> if it has a reduced expr

$s_{i_1} s_{i_2} \cdots s_{i_k}$, $i_1 > i_2 > \cdots > i_k$

A decreasing factorisation of w is a representation $w = w_k \cdots w_2 w_1$

w_k decreasing permutations, $w_i = 1$ allowed.

$\ell(w) = \ell(w_k) + \cdots + \ell(w_1)$

If $w = w_k \cdots w_1$ is a D.F. of length k

$wt(w) = (\ell(w_1), \ell(w_2), \dots \ell(w_k))$

<u>Thm</u> $\displaystyle\sum_{\substack{D.F. \\ a = w_k \cdots w_1}} z^{wt(a)} = F_w(z)$ is a symmetric polynomial. F_{w_0} is a Schur poly.

Stanley symmetric functions

<u>Thm</u> (Morse, Schilling) Decreasing factorizations have a crystal str.

This is a Stembridge crystal.

<u>Thm</u> (Stanley, ...) F_w is Schur positive. $F_w = \sum c_\lambda s_\lambda$, $c_\lambda \geq 0$.

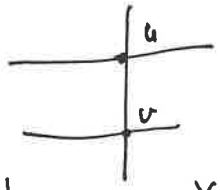
<u>Lecture 12</u>. ~~Weights~~ row transfer matrices for Γ and Δ "free-fermionic" as vertex operators

- Models related to $U_q(sl(1|1))$ "free fermionic"
- models related to $U_q(gl(2))$ "field free"

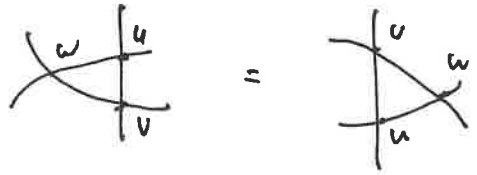
$a_1 = a_2$, $b_1 = b_2$, $c_1 = c_2$. $\dfrac{a^2+b^2-c^2}{2ab} = \underset{\text{Baxter}}{\Delta} = \frac{1}{2}(q+q^{-1})$

$a = a(u)$, $b = b(u)$, $c = c(u)$, $\Delta = \Delta(u)$

Baxter proved if $\Delta(u) = \Delta(v)$



There is a w s.t. $\Delta(w) = \Delta(u) = \Delta(v)$ s.t. YBE is satisfied



Furthermore, he extended this theory to 8 vertex model.

8VM: Heisenberg spin chains

$\longleftrightarrow$ row transfer matrices for 8VM.

Forminonic Fock space.

Dirac equation is some diff. eqn w/ solutions of various energy levels $k \in \mathbb{Z}$.

μ_k = solutions. It seemed problematic that solutions can have arbitrarily neg. energies

Solutions are fermions. Their algebra behaves like an exterior algebra. $\mu_k \wedge \mu_\ell = -\mu_\ell \wedge \mu_k$.

Pauli exclusion principle: two fermions cannot occupy same state. So there is a

Hilbert space of sol'ns of form $|j\rangle = \mu_{j_m} \wedge \mu_{j_{m-1}} \wedge \mu_{j_{m-2}} \wedge \cdots$. $j_m > j_{m-1} > j_{m-2} > \cdots$

$J_k = k$ for $k \ll 0$.

$J_k \curvearrowright F_m$, $[J_k, J_\ell] = \delta_{k+\ell}\, k I$.

A basis for F_m consists of $|\lambda\rangle = \mu_{\lambda_1+m} \wedge \mu_{\lambda_2+m-1} \wedge \cdots$ $\lambda = (\lambda_1, \lambda_2, \cdots)$ a partition.

V. Kac: Boson-Fermion correspondence relates this vector to Schur functions.

Th. $H(z) = \sum_{k=1}^{\infty} \frac{1}{k}(1-q^k) z^k J_k$, then $T(z) = e^{H(z)}$

row transf. mat.